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Technical Reproducibility Validation Report

Hipstr
Version v0.7
Y-STR

Verification date	2026-08-14
Repository commit	b2033bfbb5cf55496b776463bdf2993fa763a4be
Attestation level	**Runs + Plausible output**
Permalink	https://strhub.app/verified/hipstr-v0-7-y
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**5/5 gates passed. Runs + Plausible output**
ATTESTATION LEVEL	**Runs + Plausible output**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	Hipstr
Version	v0.7
Repository	https://github.com/tfwillems/HipSTR
Commit (pinned)	b2033bfbb5cf55496b776463bdf2993fa763a4be
Container OS	ubuntu-22.04
Marker panel	Y-STR
Verified on	2026-08-14 12:40:02 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/31801058679
Submitted by	A third party (not the tool's maintainer)

This tool was submitted for verification by somebody other than its maintainer. The maintainer took no part in the run and supplied none of what it used: the command, the environment, and any target regions were chosen by the submitter. This certificate records a run; it is not an endorsement by the tool's maintainer, and does not imply their involvement.

3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · Hipstr v0.7 · commit b2033bf
$ HipSTR \
--bams /data/in/input.bam \
--fasta /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--str-vcf /data/out/result.vcf.gz \
--min-reads 5 \
--use-unpaired \
--read-qual-trim ! \
--def-stutter-model \
--max-str-len 250
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	hipstr-v0-7-y.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	result.vcf.gz
Format	VCF
Sequence records	13
Distinct STR loci	13
Total reads	2,912
Max depth (single locus)	320

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 14 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), HG002 (Y-STR, male)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.

Ref. genome	GRCh38 / hg38
Loci tested	DYS385_2 · DYS389I · DYS389II.1 · DYS390 · DYS391 · DYS392 · DYS393 · DYS438 · DYS456 · DYS458 · DYS635 · Y-GATA-A10 · Y-GATA-H4
Regions BED	Provided by STRhub

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38), HG002 (Y-STR, male)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ **Runs end-to-end**

Hipstr v0.7 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid VCF output file with genotype calls across 13 target forensic STR loci, with a maximum read depth of 320 at Y-GATA-H4.

◆ **No bundled demo data**

Hipstr does not include its own test or demo dataset. STRhub Verified supplied Illumina BAM (hg38), HG002 (Y-STR, male) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
Y-GATA-H4	320	■■■■■■■■■■■■■■■■	**DYS456**	196	■■■■■■■■■
DYS389II.1	298	■■■■■■■■■■■■■■	**DYS391**	195	■■■■■■■■■
DYS389I	272	■■■■■■■■■■■■■	**Y-GATA-A10**	194	■■■■■■■■■
DYS635	258	■■■■■■■■■■■■	**DYS458**	193	■■■■■■■■■
DYS385_2	219	■■■■■■■■■■	**DYS392**	190	■■■■■■■■■
DYS390	218	■■■■■■■■■■	**DYS393**	161	■■■■■■■■
DYS438	198	■■■■■■■■■			